

Nim Mathematical Theory & Binary XOR Analysis

Step-by-Step Worked Derivation, Bit Parity Matrices, and State Transition Dynamics

1. Worked Example: Classic Nim [3, 4, 5]

Consider a game starting with three heaps of sizes $h_1 = 3, h_2 = 4, h_3 = 5$.

Step 1: Binary Decomposition

Convert each heap size into its standard binary column representation:

Heap	Decimal Size	Binary ($2^2 = 4$)	Binary ($2^1 = 2$)	Binary ($2^0 = 1$)
Heap 1	3	0	1	1
Heap 2	4	1	0	0
Heap 3	5	1	0	1
Nim-Sum (XOR)	2	0 (even: 2 ones)	1 (odd: 1 one)	0 (even: 2 ones)

Step 2: Identification of State Type

- The Nim-sum is $S = 3 \oplus 4 \oplus 5 = 2 \neq 0$.
- Because $S \neq 0$, this is an **N-position** (winning advantage for Player 1).
- The Most Significant Bit (MSB) of $S = 2$ is at bit position 1 (place value $2^1 = 2$).

Step 3: Deriving the Optimal Move

To force a win, the player must select a heap whose binary expansion contains a 1 at bit position 1 (MSB of S):

- Heap 1 ($3 = 011_2$):** Bit 1 is 1 $\rightarrow$ Candidate!
Target size: $h_1 \oplus S = 3 \oplus 2 = 1 < 3$.
Optimal Move: Reduce Heap 1 from 3 to 1 (remove $3 - 1 = 2$ tokens).
- Heap 2 ($4 = 100_2$):** Bit 1 is 0 $\rightarrow 4 \oplus 2 = 6 > 4$ (cannot add tokens, illegal move).
- Heap 3 ($5 = 101_2$):** Bit 1 is 0 $\rightarrow 5 \oplus 2 = 7 > 5$ (cannot add tokens, illegal move).

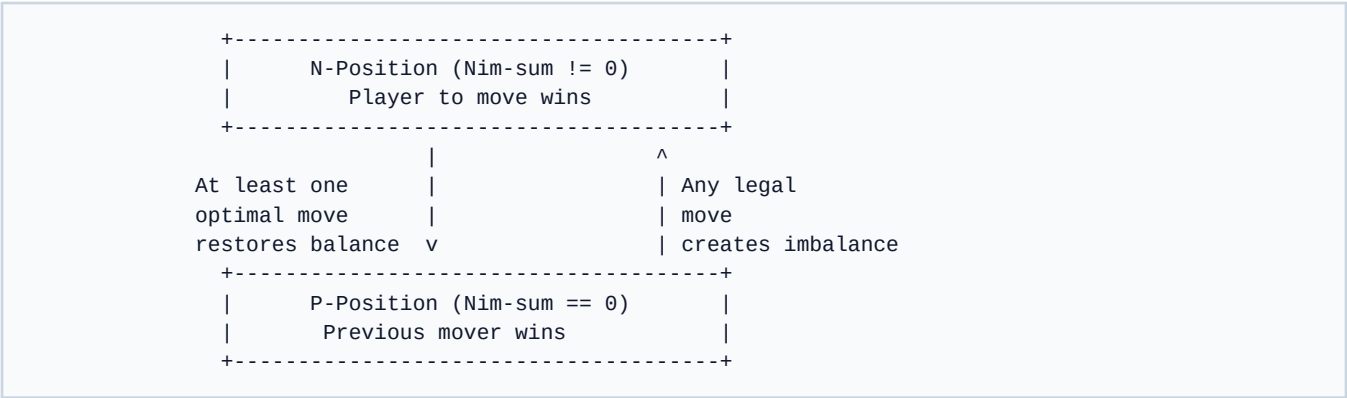
Step 4: Verifying the Resulting State

After removing 2 tokens from Heap 1, the new state is [1, 4, 5]:

Heap	Decimal Size	Binary (2^2)	Binary (2^1)	Binary (2^0)
Heap 1	1	0	0	1
Heap 2	4	1	0	0
Heap 3	5	1	0	1
New Nim-Sum (XOR)	0	0 (even)	0 (even)	0 (even)

The resulting Nim-sum is **0** (a **P-position**). The opponent receives a balanced position and has no winning move.

2. State Transition Dynamics (P and N Invariants)



The Three Fundamental Game Invariants:

- 1. Every move from a P-position leads to an N-position:
 $\forall m \in \text{LegalMoves}(P), \text{State}' \in N$
- 2. From every N-position, there exists at least one move to a P-position:
 $\exists m \in \text{LegalMoves}(N), \text{State}' \in P$
- 3. The terminal state [0, 0, ..., 0] has Nim-sum = 0, which is a P-position:
Under Normal play, the player who makes the last move wins by leaving this terminal state.